

Geometry
Unit 3
Lesson 1 Homework

Name **Answer Key** _____
Date _____
Period _____

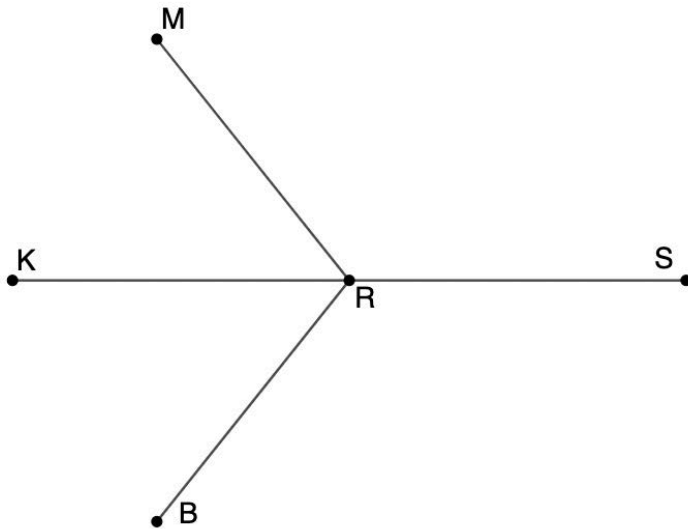
Use the phrases in the word bank to complete the proof. Some phrases may be used more than once.

Word Bank

Definition of Linear Pair	Linear Pair Postulate
Supplements of congruent angles are congruent.	Complements of congruent angles are congruent.
Angle addition postulate	Substitution property of equality
Definition of supplementary angles	Definition of complementary angles

Given: $\angle KRB \cong \angle KRM$

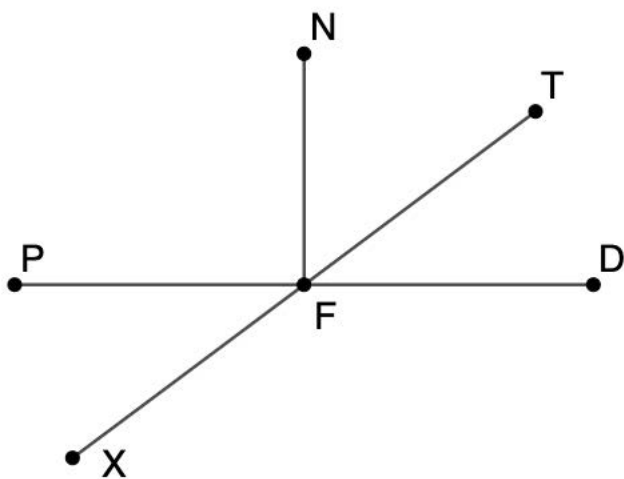
Prove: $\angle BRS \cong \angle MRS$



Complete the table using the word bank to prove $\angle BRS \cong \angle MRS$.

Statement	Reason
$\angle KRB \cong \angle KRM$	Given
$\angle KRB$ and $\angle SRB$ form a linear pair	1. Definition of Linear Pair
$\angle KRM$ and $\angle MRS$ form a linear pair	2. Definition of Linear Pair
$\angle KRB$ and $\angle SRB$ are supplementary angles	3. Linear Pair Postulate
$\angle KRM$ and $\angle MRS$ are supplementary angles	4. Linear Pair Postulate
$\angle BRS \cong \angle MRS$	5. Supplements of congruent angles are congruent.

Line segments $\overline{PD}$, $\overline{NF}$, and $\overline{TX}$ intersect at point F as shown where $m\angle PFN = 90^\circ$, $m\angle XFP = (3x - 56)^\circ$, $m\angle NFT = (4z + 7)^\circ$ and $m\angle TFD = (x + 2)^\circ$.



6. Find the value of x .

$$x = 29$$

7. Find the value of z .

$$z = 13$$

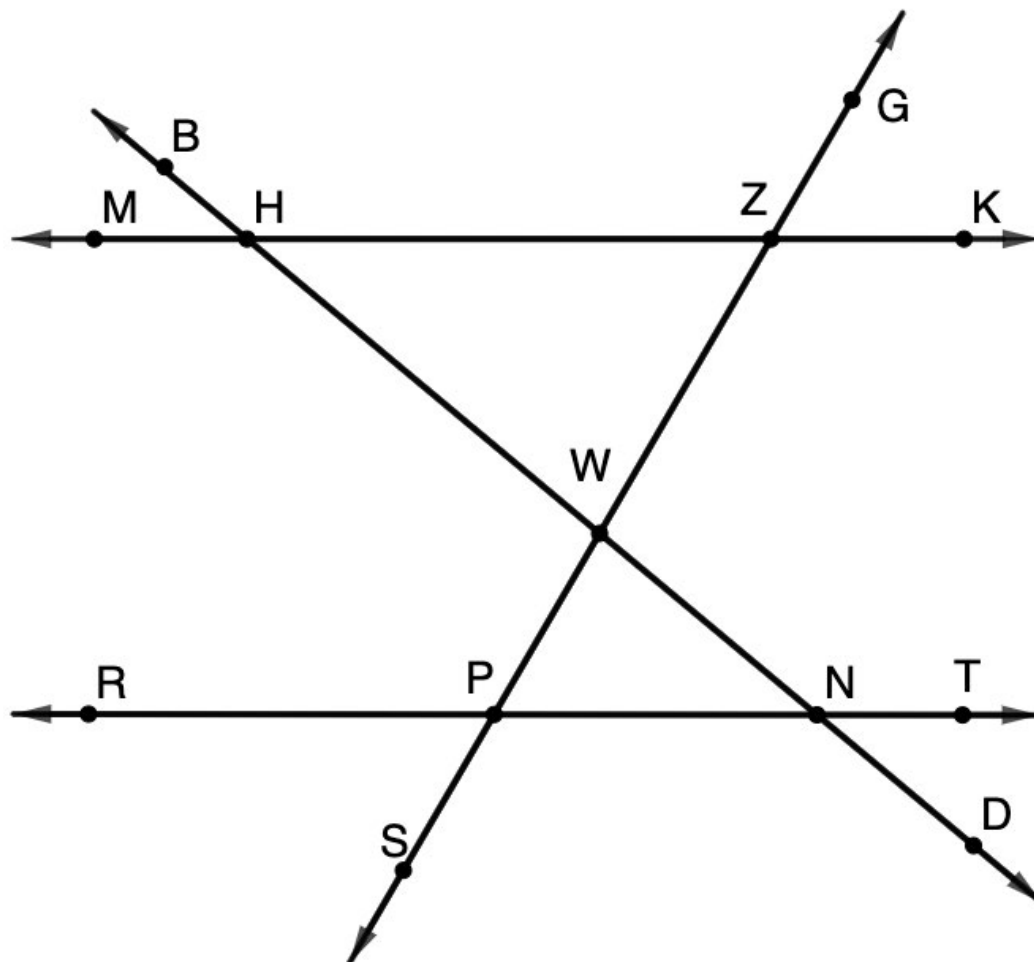
8. What is $m\angle NFT$?

$$m\angle NFT = 59^\circ$$

9. What is $m\angle XFD$?

$$m\angle XFD = 149^\circ$$

$\overleftrightarrow{MK}$, $\overleftrightarrow{RT}$, $\overleftrightarrow{BD}$ and $\overleftrightarrow{GS}$ are shown where $\overleftrightarrow{MK} \parallel \overleftrightarrow{RT}$, $\overleftrightarrow{BD} \perp \overleftrightarrow{GS}$, $m\angle MHB = 34^\circ$, $m\angle GZK = 56^\circ$, $m\angle WHZ = (9x - 2)^\circ$, and $m\angle WPN = (8y)^\circ$.



1. What is the value of x ?

$x = 4$

2. What is the value of y ?

$y = 7$

Determine the measure of the angles.

3. $m\angle WHZ$ 34°

7. $m\angle TND$ 34°

4. $m\angle WZH$ 56°

8. $m\angle HNT$ 146°

5. $m\angle HWZ$ 90°

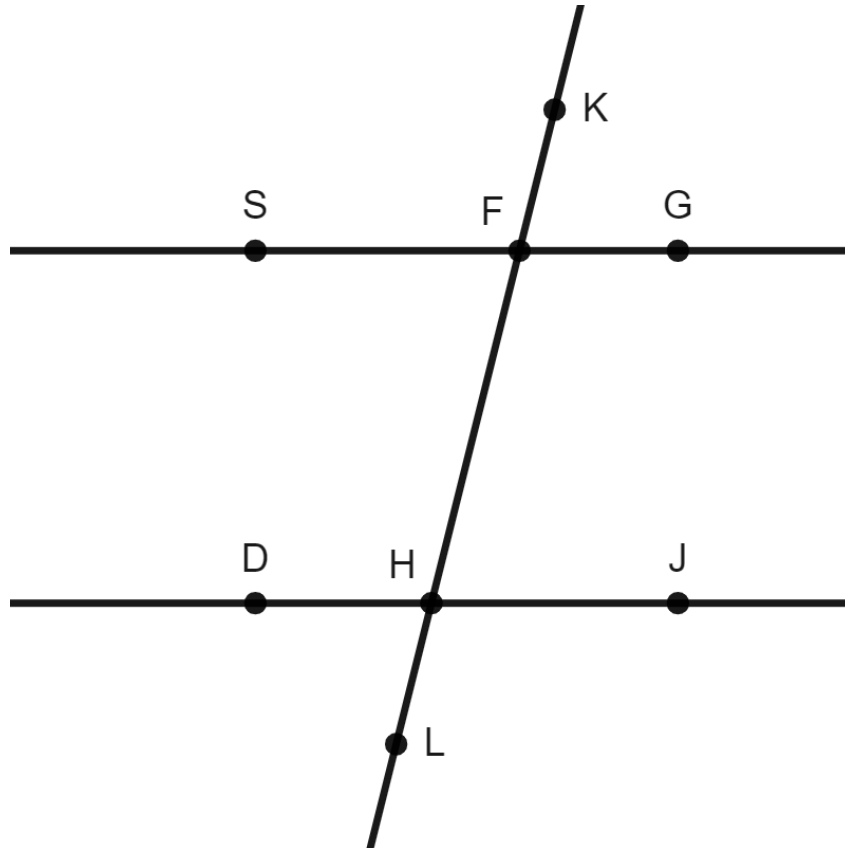
9. $m\angle PNT$ 180°

6. $m\angle RPS$ 56°

10. $m\angle SPN$ 124°

Lines $\overline{SG}$, $\overline{DJ}$, and $\overline{KL}$ are shown.

$\overline{SG} \parallel \overline{DJ}$, $m\angle FHJ = 81^\circ$



Complete the table.

Angle	Angle Measure	Relationship with $\angle FHJ$
$\angle DHL$	1. 81°	2. Vertical angle
$\angle KFG$	3. 81°	4. Corresponding angle pair
$\angle SFH$	5. 81°	6. Alternate interior angle pair

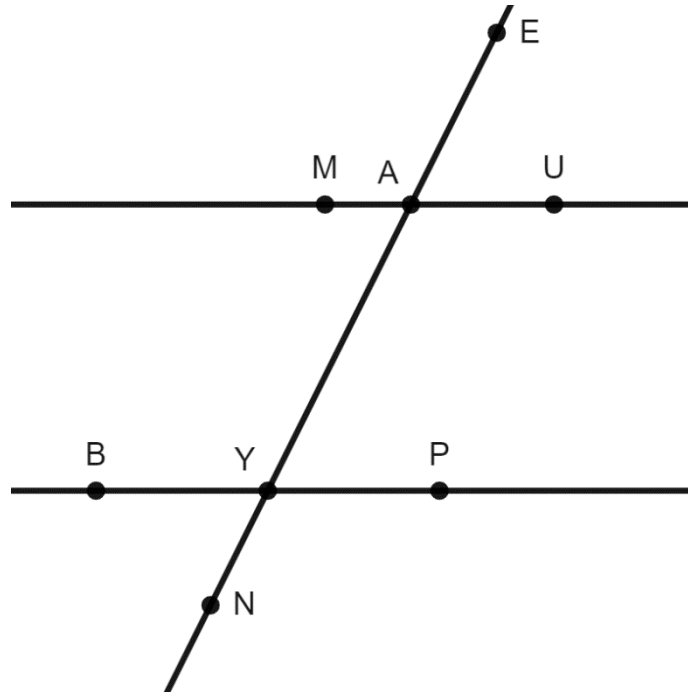
$\angle FHJ$ and $\angle FHD$ are supplementary, so $m\angle FHD = 99^\circ$.

Complete the table.

Angle	Angle Measure	Relationship with $\angle FHD$
$\angle LHJ$	7. 99°	8. Vertical angle pair
$\angle KFS$	9. 99°	10. Corresponding angle pair

Lines $\overline{MU}$, $\overline{BP}$, and $\overline{NE}$ are shown.

$\overline{MU} \parallel \overline{BP}$, $m\angle EAU = 63^\circ$



Identify the relationship between each pair of angles.

Angles	Congruent or Supplementary?
$\angle EAU$ and $\angle YAU$	1. Supplementary
$\angle EAU$ and $\angle PYN$	2. Supplementary
$\angle EAU$ and $\angle MAE$	3. Supplementary
$\angle EAU$ and $\angle MAY$	4. Congruent

Complete the table according to the diagram.

Angle	Angle Measure	Relationship with $\angle MAY$
$\angle BYA$	5. 117°	6. Consecutive Interior Angles
$\angle MAE$	7. 117°	8. Linear Pair

9. If $m\angle EAU = (7x - 14)^\circ$ and $m\angle PYN = (11x - 4)^\circ$, what is the value of x ?

$$7x - 14 + 11x - 4 = 180$$

$$18x - 18 = 180$$

$$18x = 198$$

$$x = 11$$

10. If $m\angle MAY = (4w - 5)^\circ$ and $m\angle BYA = (6w + 15)^\circ$, what is the value of w ?

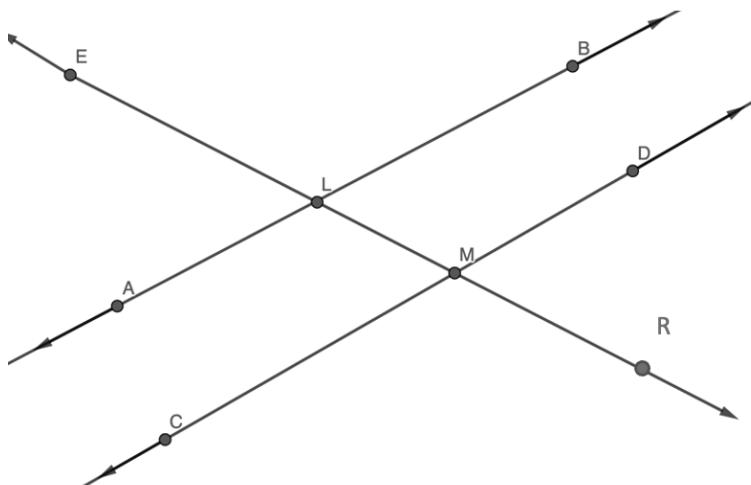
$$4w - 5 + 6w + 15 = 180$$

$$10w + 10 = 180$$

$$10w = 170$$

$$w = 17$$

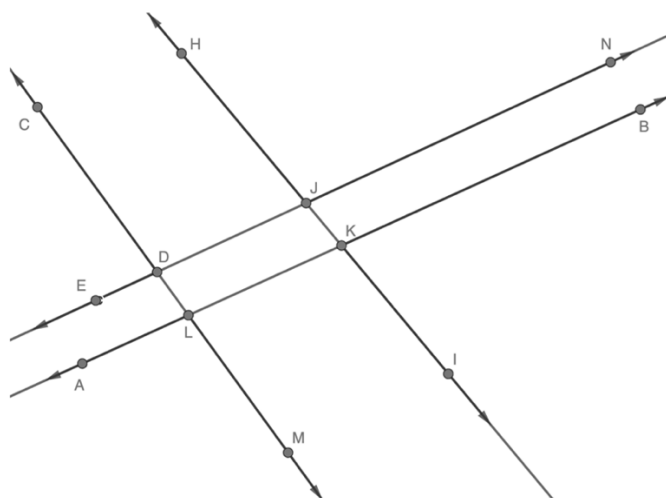
1. Given $\overleftrightarrow{AB}$, $\overleftrightarrow{CD}$, and $\overleftrightarrow{ER}$. $\overleftrightarrow{ER}$ intersects $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$.



Determine if the given condition could be used to justify that $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$. Then, justify your answer.

Condition	Is $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$?	Justification
$\angle ELA \cong \angle DMR$	☒ Yes ☐ No	Converse alternate exterior angles
$\angle ALM \cong \angle BLE$	☐ Yes ☒ No	Vertical angles do not prove lines parallel
$m\angle CMR + m\angle RMD = 180^\circ$	☐ Yes ☒ No	Linear pair does not prove lines parallel
$\angle ELB \cong \angle LMD$	☒ Yes ☐ No	Corresponding angles
$\angle ELB \cong \angle RMD$	☐ Yes ☒ No	Congruent consecutive exterior angles do not always prove parallel lines
If $m\angle ELB = 102^\circ$, then $m\angle ELA = 78^\circ$	☐ Yes ☒ No	Linear pair does not prove lines parallel
If $m\angle RMC = 91^\circ$, then $m\angle ALM = 89^\circ$	☒ Yes ☐ No	Corresponding angles are congruent in parallel lines

2. $\overleftrightarrow{CM}$, $\overleftrightarrow{HI}$, $\overleftrightarrow{EN}$, and $\overleftrightarrow{AB}$ are shown where $m\angle LKI = 94^\circ$.



Determine if the given conditions could be used to justify that $\overleftrightarrow{EN} \parallel \overleftrightarrow{AB}$ or $\overleftrightarrow{CM} \parallel \overleftrightarrow{HI}$. Then justify your answer.

Condition	Is $\overleftrightarrow{EN} \parallel \overleftrightarrow{AB}$ or $\overleftrightarrow{CM} \parallel \overleftrightarrow{HI}$?	Justification
$\angle CDJ \cong \angle KLM$	☐ Yes ☒ No	If two lines intersected by a transversal where consecutive exterior angles are congruent, does not mean the lines are parallel.
$\angle CDE \cong \angle MLA$	☐ Yes ☒ No	If two lines intersected by a transversal where consecutive exterior angles are congruent, does not mean the lines are parallel.
$m\angle EDL + m\angle NJK = 180^\circ$	☒ Yes ☐ No	If two lines intersected by a transversal where consecutive exterior angles are supplementary, then the lines are parallel.
$\angle KLM \cong \angle HJN$	☐ Yes ☒ No	The two angles do not share two lines and a transversal.
$\angle JKL \cong \angle BKI$	☐ Yes ☒ No	Congruent vertical angles do not prove parallel lines.
If $\angle BKI = 86^\circ$, then $m\angle BKI + m\angle LKI = 180^\circ$	☐ Yes ☒ No	Angles in a linear pair do not prove parallel lines.

$\overleftrightarrow{MN} \parallel \overleftrightarrow{PH}$, where $\overleftrightarrow{RV}$ is a transversal with
 $m\angle RSM = (12x - 7)^\circ$, and $m\angle PTV = (4x - 5)^\circ$ as shown.

1. Determine the value of x .

$x = 12$

2. What is $m\angle RSM$?

137°

3. What is $m\angle PTV$?

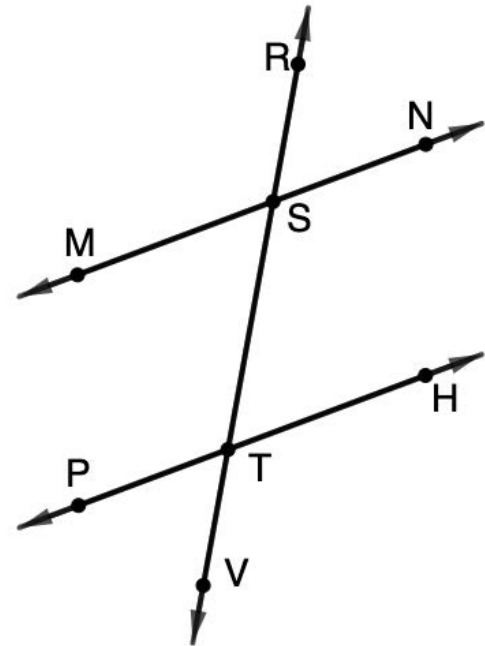
43°

4. List all the angles congruent to $\angle RSM$.

$\angle RSM \cong \angle NST \cong \angle HTV \cong \angle PTS$

5. List all the angles congruent to $\angle PTV$.

$\angle PTV \cong \angle STH \cong \angle RSN \cong \angle MST$



$\overleftrightarrow{MN} \parallel \overleftrightarrow{PH}$, where $\overleftrightarrow{RV}$ is a transversal with $m\angle RSN = (3x - 17)^\circ$ and $m\angle STP = (7x - 3)^\circ$ as shown.

6. Determine the value of x .

$$x = 20$$

7. What is $m\angle RSN$?

$$43^\circ$$

8. What is $m\angle STP$?

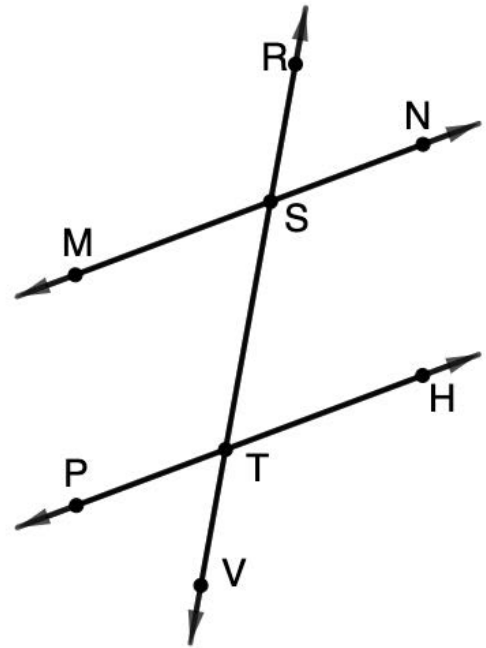
$$137^\circ$$

9. What is $m\angle RSM$?

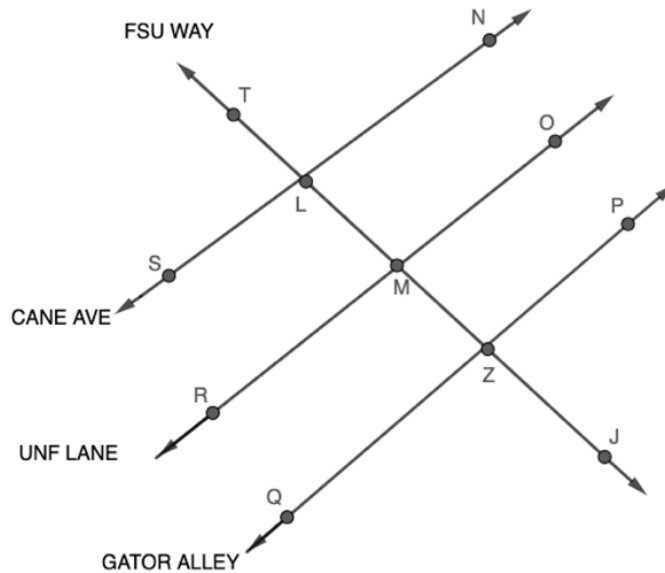
$$137^\circ$$

10. What is $m\angle HTS$?

$$43^\circ$$



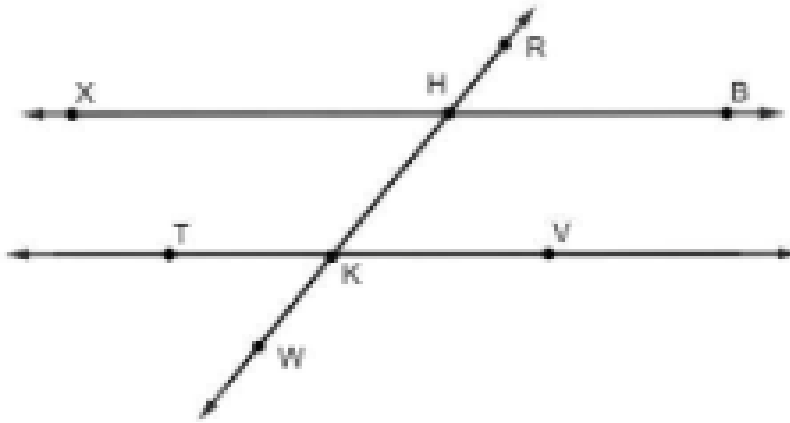
Use the following information for questions 1-3. FSU Way, a transversal, TJ , crosses the streets Cane Avenue, UNF Lane, and Gator Alley.



Determine the value of the variables and each angle measure given Cane Avenue, UNF Lane, and Gator Alley are parallel to each other.

- $\angle TLN = (3x - 18)^\circ$, $\angle MZQ = (5x + 14)^\circ$, and $\angle NLM = y^\circ$
 $3x - 18 + 5x + 14 = 180$
 $8x - 4 = 180$
 $8x = 184$
 $x = 23$
 $3(23) - 18 = 69 - 18 = 51^\circ = m\angle TLN$
 $5(23) + 14 = 115 + 14$
 $= 129^\circ = m\angle MZQ$
 $180 - 51 = 129 = y$
 $129^\circ = m\angle NLM$
- $\angle JZQ = (9x + 9)^\circ$ and $\angle TLN = (8x + 19)^\circ$
 $9x + 9 = 8x + 19$
 $x = 10$
 $9(10) + 9 = 90 + 9 = 99^\circ = m\angle JZQ$
 $8(10) + 19 = 80 + 19 = 99^\circ = m\angle TLN$
- $\angle ZMR = (x + 14)^\circ$, $\angle LMO = (2x - 6)^\circ$, and $\angle LMR = y^\circ$
 $x + 14 = 2x - 6$
 $20 = x$
 $20 + 14 = 34^\circ = m\angle ZMR$
 $2(20) - 6 = 40 - 6 = 34^\circ = m\angle LMO$
 $34 + y = 180$
 $y = 146$
 $m\angle LMR = 146^\circ$

Use the following information for questions 4-7. Given $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown. $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ where $\overleftrightarrow{RW}$ is a transversal.



Determine the value of the variables and each angle measure.

4. $m\angle TKW = (8x - 20)^\circ$ and $m\angle XHK = (3x + 2)^\circ$

$$8x - 20 = 3x + 2$$

$$5x = 22$$

$$x = 4\frac{2}{5}$$

$$8\left(\frac{22}{5}\right) - 20 = 35\frac{1}{5} - 20 = 15\frac{1}{5}^\circ = m\angle TKW$$

$$3\left(\frac{22}{5}\right) + 2 = 13\frac{1}{5} + 2 = 15\frac{1}{5}^\circ = m\angle XHK$$

5. $m\angle RHB = (2x - 20)^\circ$ and $m\angle XHK = 44^\circ$

$$2x - 20 = 44$$

$$2x = 64$$

$$x = 32$$

$$m\angle RHB = 44^\circ$$

6. $m\angle HKT = (3y - 18)^\circ$, $m\angle VKW = (8x + 12)^\circ$, and $m\angle RHB = (2x + 18)^\circ$

$$8x + 12 + 2x + 18 = 180$$

$$10x + 30 = 180$$

$$10x = 150$$

$$x = 15$$

$$2(15) + 18 = 30 + 18 = 48 =$$

$$m\angle RHB = 48^\circ$$

$$3y - 18 = 8x + 12$$

$$3y - 18 = 8(15) + 12$$

$$3y - 18 = 120 + 12$$

$$3y - 18 = 132$$

$$3y = 150$$

$$y = 50$$

$$m\angle VKW = 132^\circ$$

$$3(50) - 18 = 150 - 18 = 132^\circ = m\angle HKT$$

7. $m\angle BHK = (8x + 12)^\circ$ and $m\angle VKW = 140^\circ$

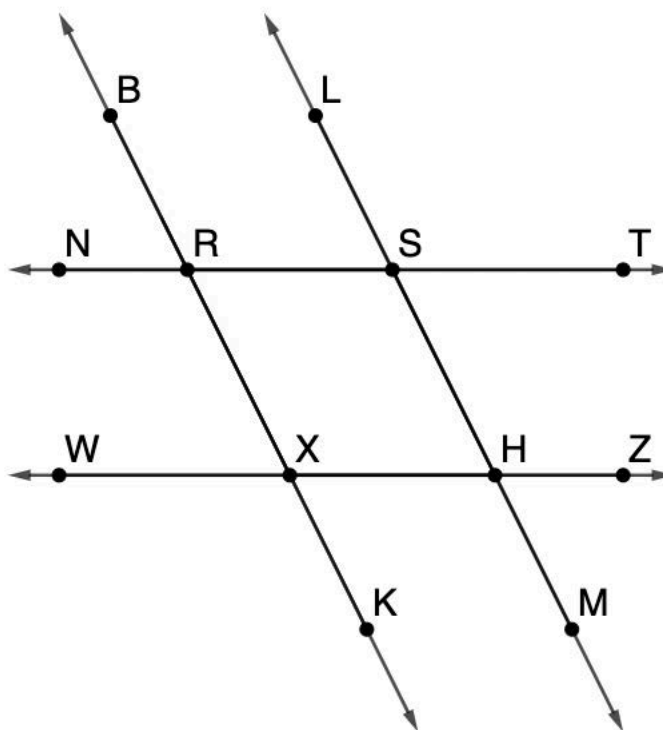
$$8x + 12 = 140$$

$$8x = 128$$

$$x = 16$$

$$8(16) + 12 = 128 + 12 = 140^\circ = m\angle BHK$$

Lines $\overleftrightarrow{BK}$, $\overleftrightarrow{LM}$, $\overleftrightarrow{NT}$ and $\overleftrightarrow{WZ}$ are shown.



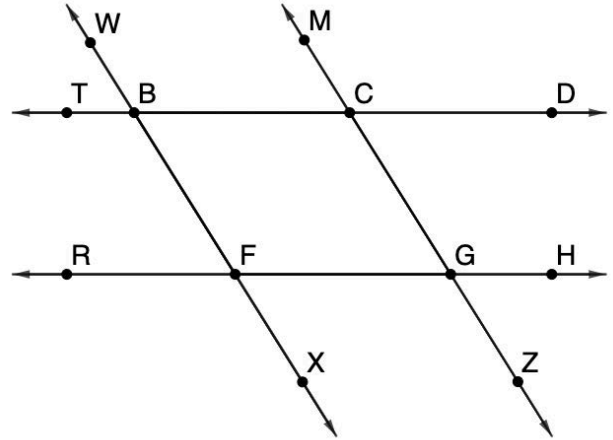
Circle the contradictory piece of information in each row.

	Item 1	Item 2	Item 3	Item 4
1.	$\angle RSL \cong \angle XHM$	$\angle BRN \cong \angle XRS$	$\angle WXK$ and $\angle NRB$ are supplementary	$\overleftrightarrow{NT} \parallel \overleftrightarrow{WZ}$
2.	$\overleftrightarrow{BK} \parallel \overleftrightarrow{LM}$	$\angle LST$ and $\angle NRX$ are supplementary	$m\angle LST = 120^\circ$	$\angle WXK \cong \angle SHZ$
3.	$\angle LST$ and $\angle WXR$ are supplementary	$\overleftrightarrow{NT} \parallel \overleftrightarrow{WZ}$	$\overleftrightarrow{BK} \parallel \overleftrightarrow{LM}$	$m\angle BRS = 195^\circ$
4.	$\overleftrightarrow{NT} \parallel \overleftrightarrow{WZ}$	$\angle RSL \cong \angle HST$	$\overleftrightarrow{BK} \nparallel \overleftrightarrow{LM}$	$\angle NRB \cong \angle MHZ$
5.	$\angle NRX \cong \angle LST$	$\angle NRB$ and $\angle MHZ$ are supplementary	$\angle LST$ and $\angle MHZ$ are supplementary	$\angle RSH \cong \angle HXR$

Choose from the bank of statements and reasons to complete the paragraph proof.

Given: $\angle MCD \cong \angle CGH$

Prove: $\angle CBF \cong \angle XFG$



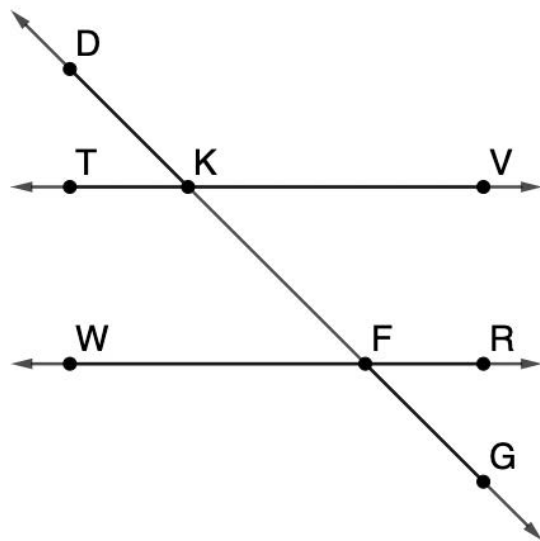
It is given that (6) $\angle MCD \cong \angle CGH$. Since the (7) corresponding angles are congruent, it must be true that (8) $\overrightarrow{TD} \parallel \overrightarrow{RH}$. If two parallel lines are intersected by a transversal, consecutive angles are supplementary, then it must be true that (9) $\angle CBF$ and $\angle BFG$ are supplementary. Angles $\angle BFG$ and $\angle XFG$ are supplementary because they form (10) a linear pair. Since supplements of the same angle are congruent, it must be true that $\angle CBF \cong \angle XFG$.

Statements and Reasons	
$\angle MCD \cong \angle CGH$	vertical angles
$\angle MCD \cong \angle RFX$	a linear pair
$\angle FGZ \cong \angle ZGH$	consecutive angles are supplementary
$\overrightarrow{TD} \parallel \overrightarrow{RH}$	corresponding angles are congruent
$\overrightarrow{WB} \parallel \overrightarrow{MZ}$	alternate interior angles are congruent
$\angle TBW$ and $\angle MCD$	converse of the corresponding angle theorem
$\angle CBF$ and $\angle BFG$	
$\angle WBC$ and $\angle XFG$	

Complete the two-column proof.

Given: $\angle TKF \cong \angle KFR$

Prove: $\angle VKD \cong \angle WFG$



Statement	Reason
$\angle TKF \cong \angle KFR$	Given
1. $\overleftrightarrow{TV} \parallel \overleftrightarrow{WR}$	2. If alternate interior angles are congruent, then the two lines that are intersected by a transversal are parallel.
3. $\angle VKD \cong \angle WFG$	4. If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.

Write the statements and reasons in the correct order in the two-column proof.

$$\angle TKD \cong \angle GFR$$

Supplements of congruent angles are congruent.

$$\angle VKD \cong \angle WFG$$

If two angles form a linear pair, then they are supplementary

$$\overleftrightarrow{TV} \parallel \overleftrightarrow{WR}$$

If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.

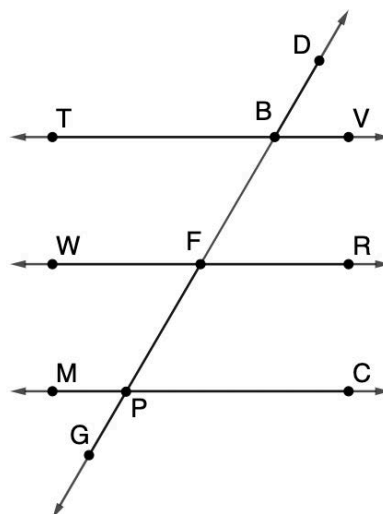
$\angle GFR$ and $\angle WFG$ are supplementary

If alternate interior angles are congruent, then the two lines that are intersected by a transversal are parallel.

Complete the two-column proof.

Given: $\angle DBV \cong \angle MPG$ and $\angle TBD \cong \angle WFB$

Prove: $\angle RFP \cong \angle MPF$



Statement	Reason
$\angle DBV \cong \angle MPG$	Given
5. $\overleftrightarrow{TV} \parallel \overleftrightarrow{MC}$	5. If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.
6. $\angle TBD \cong \angle WFB$	6. Given
7. $\overleftrightarrow{TV} \parallel \overleftrightarrow{WR}$	7. If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.
8. $\angle PFR \cong \angle TBD$	8. If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.
9. $\angle TBD \cong \angle MPF$	9. If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
10. $\angle RFP \cong \angle MPF$	10. Transitive Property of Congruence

Write the statements and reasons in the correct order in the two-column proof.

$\angle RFP \cong \angle MPF$ Given

$\angle TBD \cong \angle WFB$ Transitive Property of Congruence

$\angle PFR \cong \angle TBD$ If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.

$\angle TBD \cong \angle MPF$ If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.

$\overleftrightarrow{TV} \parallel \overleftrightarrow{MC}$ If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.

$\overleftrightarrow{TV} \parallel \overleftrightarrow{WR}$ If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.